



Binny & Bloom





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Inventory Tracking
Smart Recipes



Donation Advisor
Compost Guide

Team Members: Abigail - Taja - Luis - Emma - Jay

Why We Built This

- Food waste is a major environmental and financial issue
- People forget what they bought
- Items expire unnoticed
- Recipes often require ingredients you *don't* have
- Donation rules are confusing
- Composting is even more confusing





Meet Binny & Bloom



SMART INVENTORY
TRACKING



EXPIRY WARNINGS
(4-DAY + 1-DAY
URGENT)



AI-GENERATED
RECIPES USING
ONLY WHAT YOU
HAVE



DONATION
ADVISOR FOR
SAFE, UNOPENED
ITEMS



COMPOSTING
GUIDE FOR
EXPIRED ITEMS



SUSTAINABILITY
SCORING SYSTEM



FRIENDLY UI



How Everything Works Together

- Flask backend
- SQLite database
- AI integration via OpenRouter
- Custom validator for safe recipes
- Front-end templates + CSS
- CLI version for terminal users

• User -> Flask -> Database -> AI -> Validator -> UI



Smart Inventory + Player Tracking

- Automatic flagging for raw meat, perishables
- Covers donation eligibility or compostability
- Scoring system that tracks value saved by cooking instead of wasting
- Expiry detection that triggers warnings on dashboard

AI That Actually Understands Your Inventory

- Strict JSON schema
- No invented ingredients
- No expired items
- Respects allergies + dietary needs
- Validated with custom Python logic

← Four modes:

1. Chat
2. Recipe
3. Donation
4. Decomposition

Keeping Recipes Safe & Realistic



- **Logistics:**
 - Checks ingredient availability
 - Converts units (g/ml/count)
- **Safety:**
 - Rejects expired items
 - Rejects allergens
 - Rejects dietary conflicts
- **Precision:**
 - Ensures realistic cooking amounts



Your Sustainability Command Center

$$\text{Efficiency Score \%} = \frac{\text{Total value of items ate/donated/composted}}{\text{Total purchase value of items}} \times 100$$

- Sustainability tiers:
 - Efficiency ≥ 90 : Eco-Legend
 - Efficiency ≥ 75 : Eco-Warrior
 - Efficiency ≥ 50 : Apprentice
 - Else: Novice
- Expiry warnings
- Saved recipes

Keeping You Ahead Of Spoilage



- ▶ 4-day warning
- ▶ 1-day urgent warning
- ▶ Dashboard flash messages
- ▶ Items sorted into:
 - ▶ Expired
 - ▶ About to expire
 - ▶ Fresh
- ▶ **Visual Suggestions:**
 - ▶ 🚨 and 🚩 icons
 - ▶ Color-coded categories

What We'll Show You

- Login
- Inventory
- Dashboard
- AI Menu
- Generating a recipe

Individual Contributions



Abby



Taja



Emma



Jay



Luis

Questions?



Symbols assigned



Jay



Emma



Taja



Abby



Luis